2-D airfoil rollout30 - literature ROM comparison

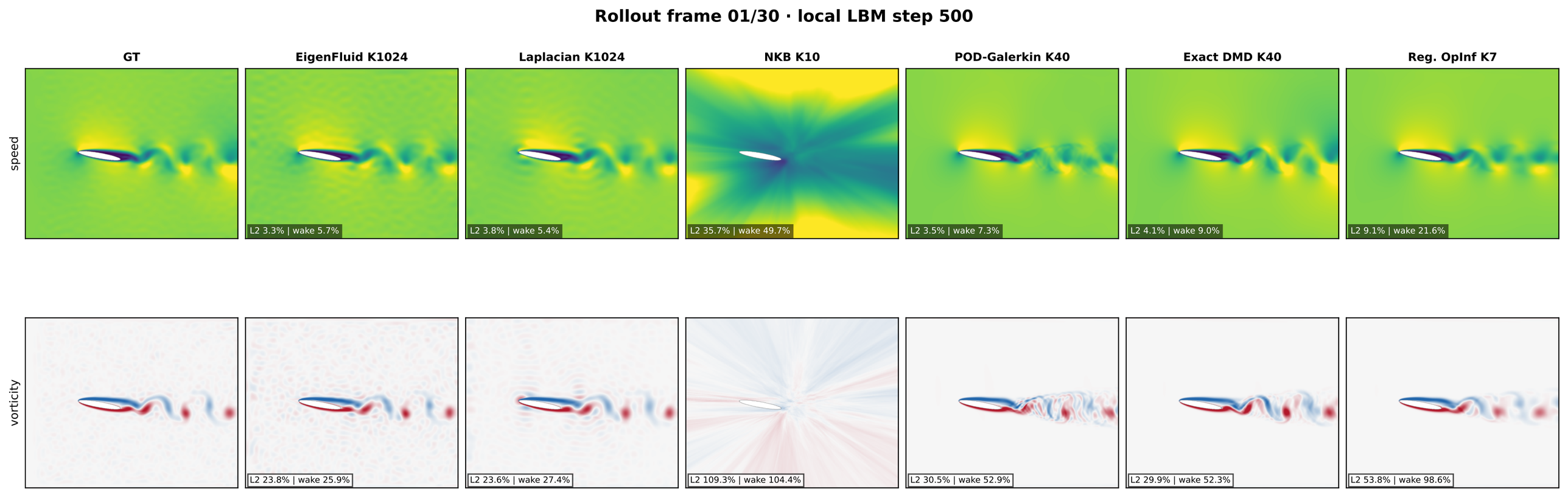
Constructed D2Q9 TRT-LBM GT, Re=1500, beta=0.50, alpha=+10 deg, 500 lattice steps per frame

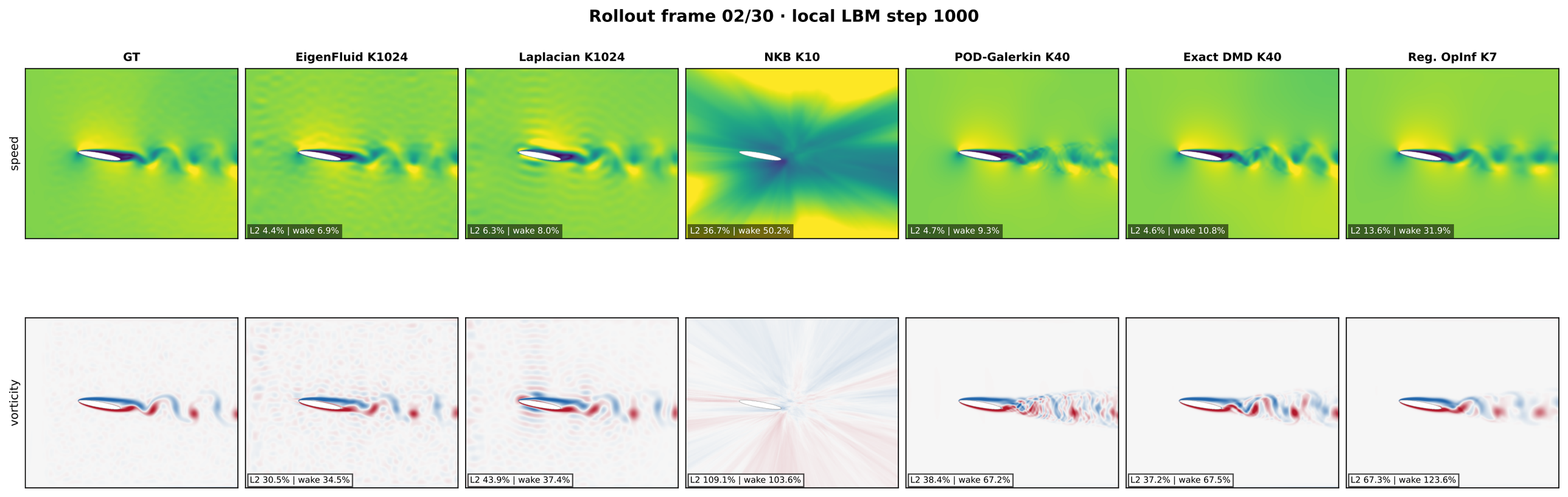
Method	velocity full	velocity wake	vorticity full	vorticity wake
EigenFluid K1024	7.39% / 9.35%	16.14% / 21.89%	54.17% / 64.51%	78.58% / 98.25%
Laplacian K1024	17.53% / 20.17%	26.76% / 30.15%	91.85% / 97.38%	100.76% / 107.55%
NKB K10	84.07% / 181.61%	58.93% / 90.03%	120.92% / 151.95%	111.89% / 134.05%
POD-Galerkin K40	5.76% / 5.95%	12.66% / 13.24%	44.51% / 44.30%	78.27% / 76.56%
Exact DMD K40	5.73% / 6.35%	13.22% / 13.38%	39.91% / 36.83%	70.62% / 61.95%
Reg. OpInf K7	9.62% / 8.33%	22.86% / 18.53%	53.92% / 39.57%	96.65% / 68.16%

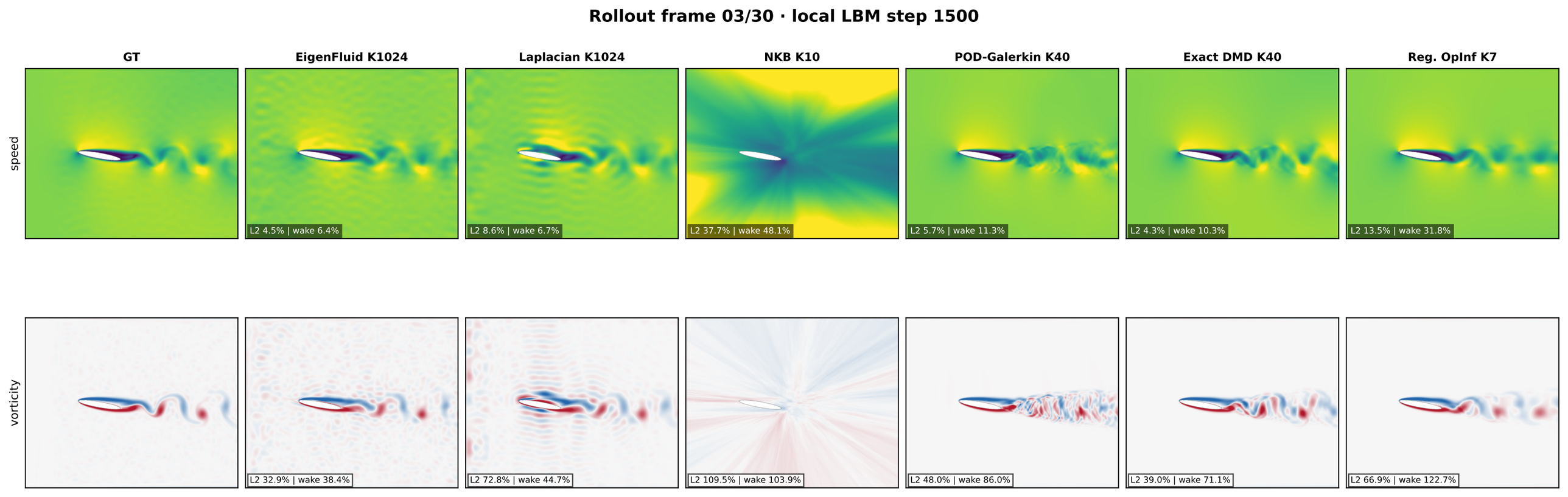
Each entry is 30-frame mean / final relative L2. Training uses 41 GT snapshots for the three literature ROMs; the 30-frame evaluation trajectory is independent and initialized from the same reconstructed velocity for every method.

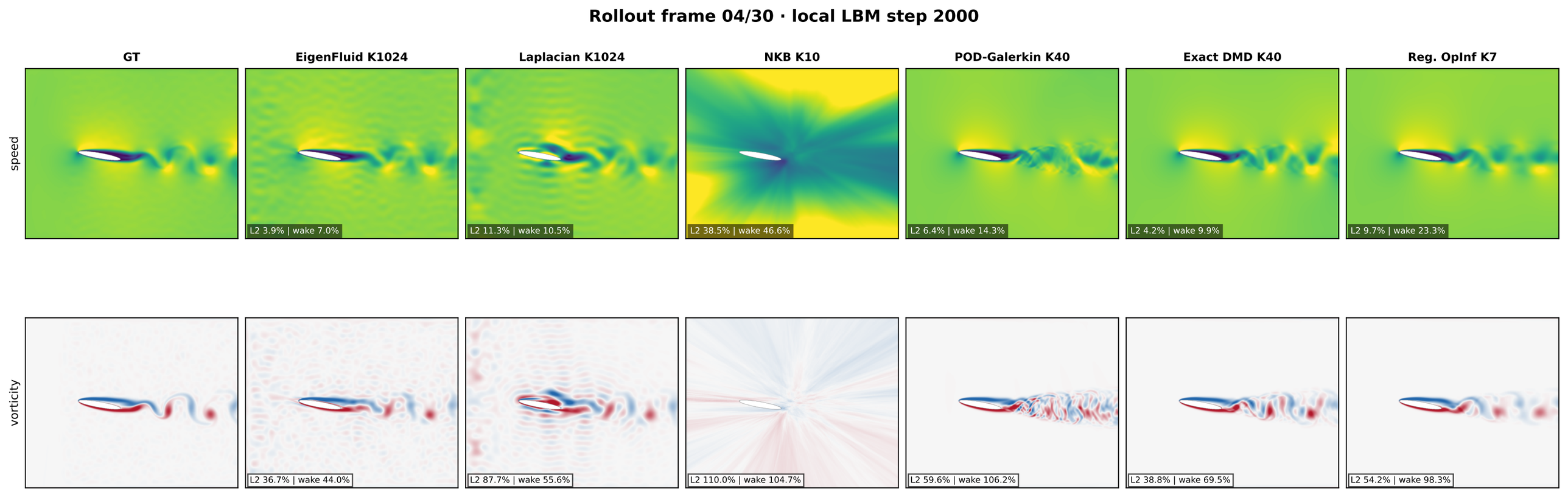
NKB follows camera-ready Eq. (8): backward trace, Euclidean closest-domain projection, source-velocity copy, and least-squares projection. POD and DMD use the float64-resolved snapshot rank; regularized OpInf uses the largest resolved quadratic feature rank and training-trajectory stability selection.

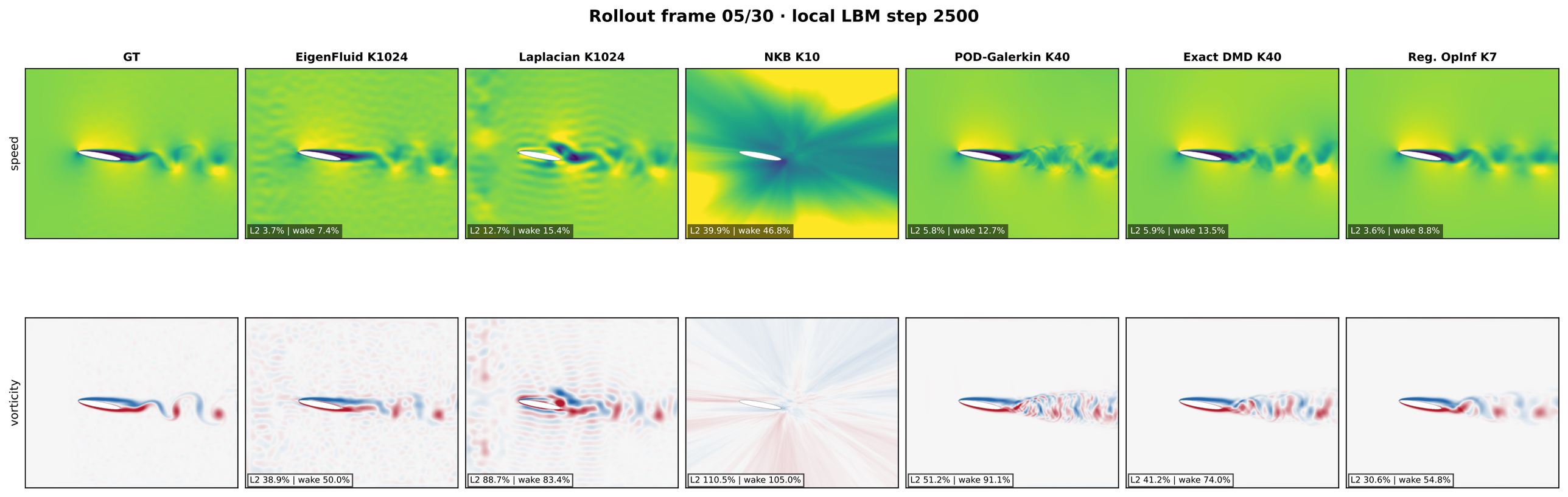
Primary references: Noack et al. (POD-Galerkin); Schmid 2010 (DMD); McQuarrie, Huang & Willcox 2021 (regularized Operator Inference); Liu et al. 2025 (Neural Kinematic Bases).

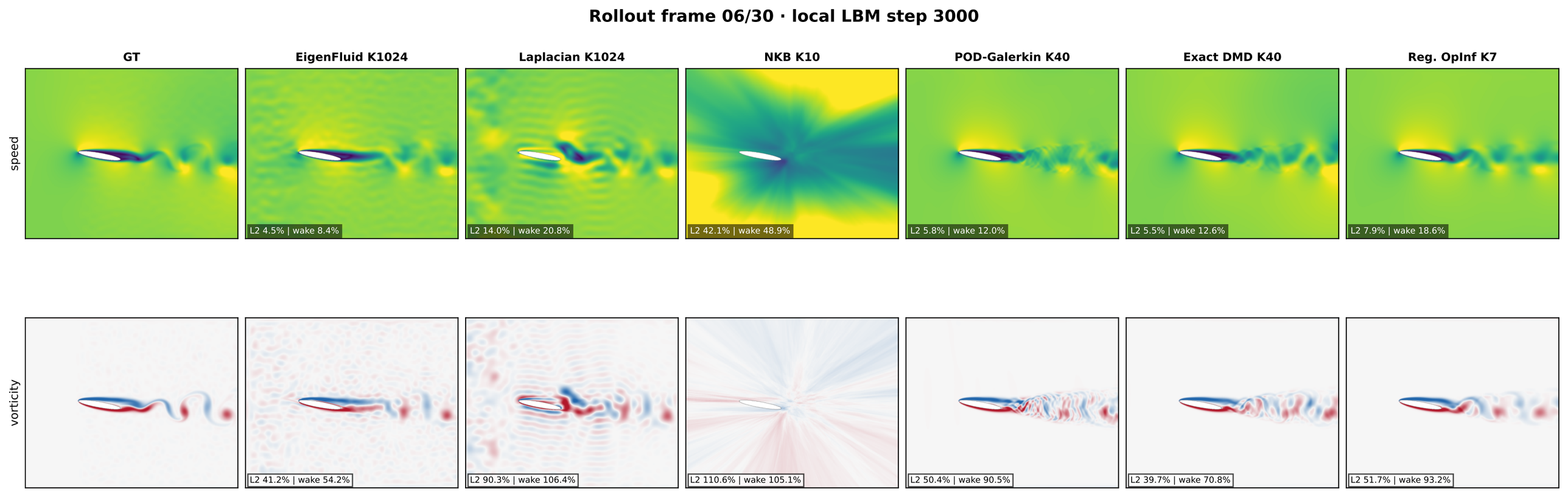

Rollout frame 01/30 · local LBM step 500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 3.3% | wake 5.7%
L2 3.8% | wake 5.4%
L2 35.7% | wake 49.7%
L2 3.5% | wake 7.3%
L2 4.1% | wake 9.0%
L2 9.1% | wake 21.6%
L2 23.8% | wake 25.9%
L2 23.6% | wake 27.4%
L2 109.3% | wake 104.4%
L2 30.5% | wake 52.9%
L2 29.9% | wake 52.3%
L2 53.8% | wake 98.6%

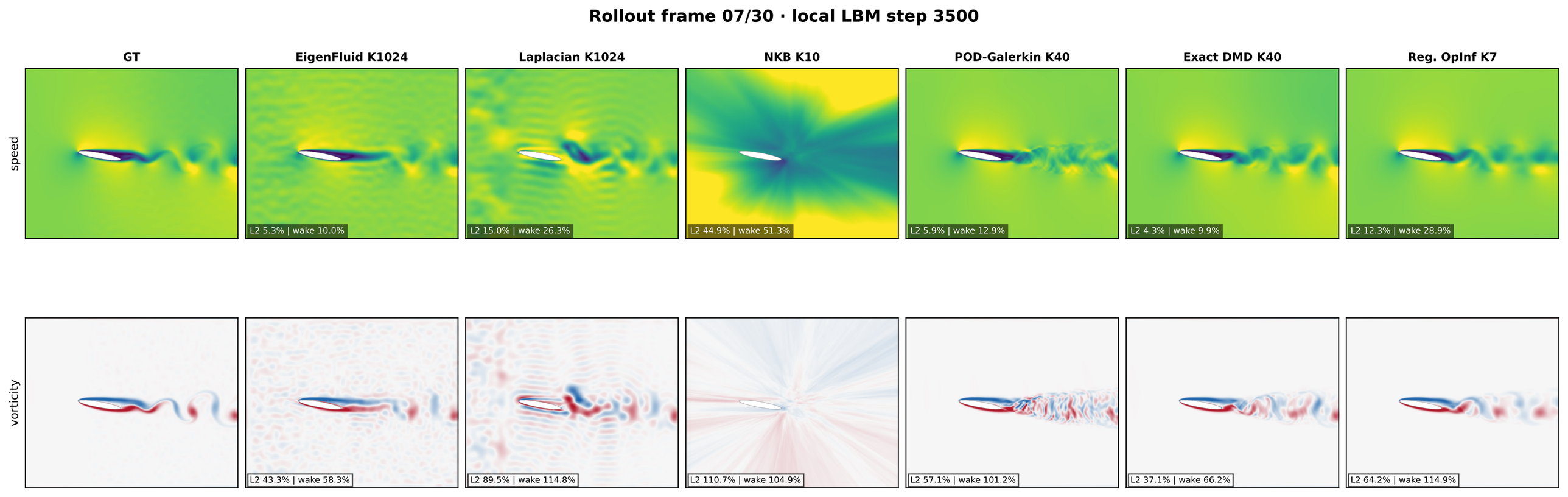

Rollout frame 02/30 · local LBM step 1000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 4.4% | wake 6.9%
L2 6.3% | wake 8.0%
L2 36.7% | wake 50.2%
L2 4.7% | wake 9.3%
L2 4.6% | wake 10.8%
L2 13.6% | wake 31.9%
L2 30.5% | wake 34.5%
L2 43.9% | wake 37.4%
L2 109.1% | wake 103.6%
L2 38.4% | wake 67.2%
L2 37.2% | wake 67.5%
L2 67.3% | wake 123.6%

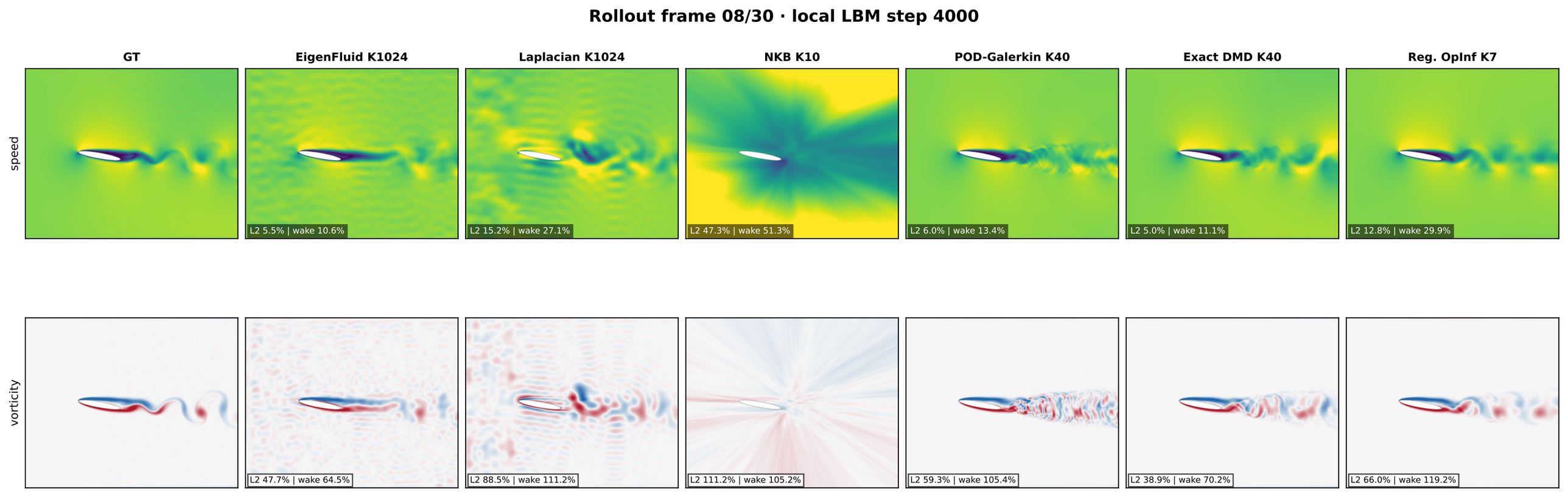

Rollout frame 03/30 · local LBM step 1500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 4.5% | wake 6.4%
L2 8.6% | wake 6.7%
L2 37.7% | wake 48.1%
L2 5.7% | wake 11.3%
L2 4.3% | wake 10.3%
L2 13.5% | wake 31.8%
L2 32.9% | wake 38.4%
L2 72.8% | wake 44.7%
L2 109.5% | wake 103.9%
L2 48.0% | wake 86.0%
L2 39.0% | wake 71.1%
L2 66.9% | wake 122.7%

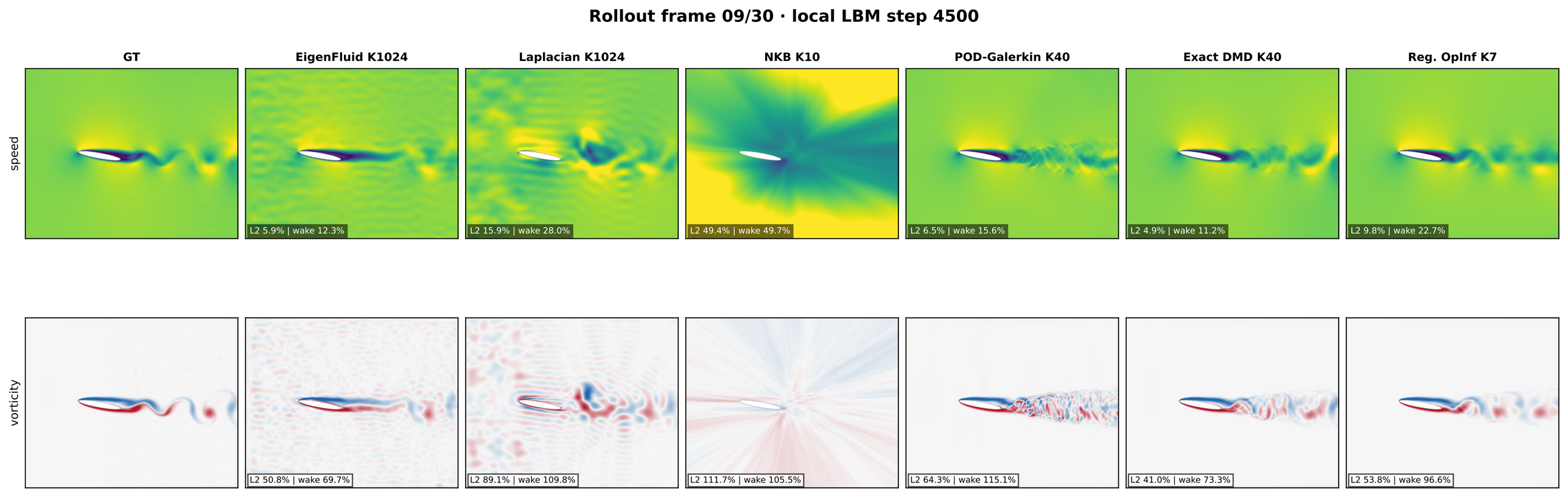

Rollout frame 04/30 · local LBM step 2000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 3.9% | wake 7.0%
L2 11.3% | wake 10.5%
L2 38.5% | wake 46.6%
L2 6.4% | wake 14.3%
L2 4.2% | wake 9.9%
L2 9.7% | wake 23.3%
L2 36.7% | wake 44.0%
L2 87.7% | wake 55.6%
L2 110.0% | wake 104.7%
L2 59.6% | wake 106.2%
L2 38.8% | wake 69.5%
L2 54.2% | wake 98.3%

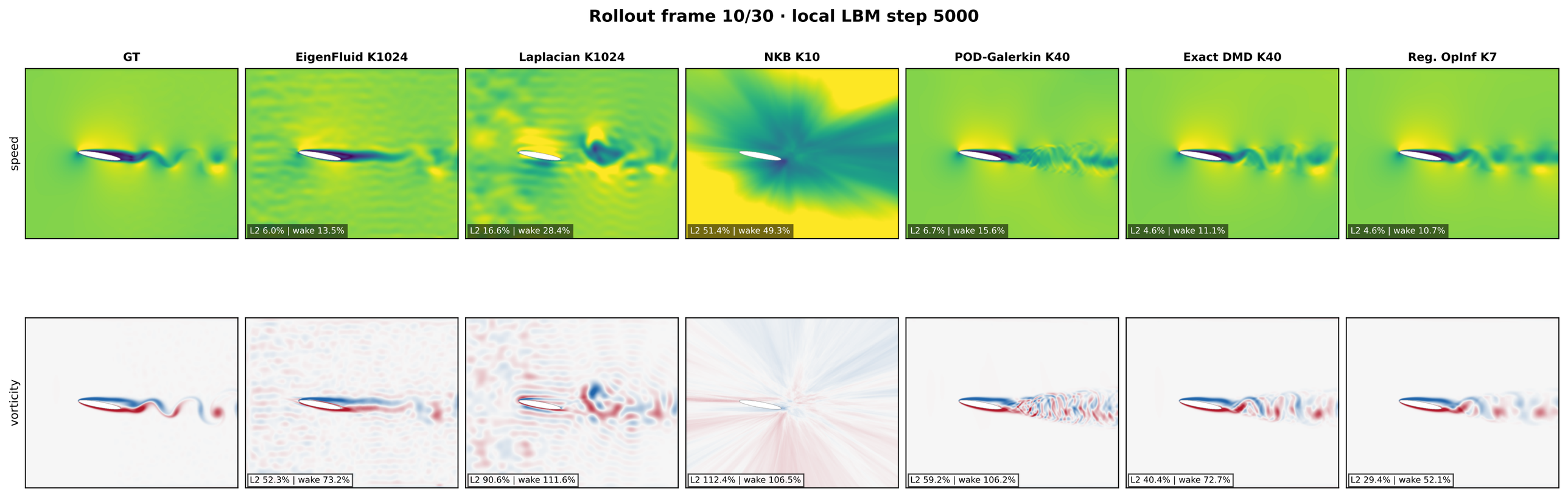

Rollout frame 05/30 · local LBM step 2500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 3.7% | wake 7.4%
L2 12.7% | wake 15.4%
L2 39.9% | wake 46.8%
L2 5.8% | wake 12.7%
L2 5.9% | wake 13.5%
L2 3.6% | wake 8.8%
L2 38.9% | wake 50.0%
L2 88.7% | wake 83.4%
L2 110.5% | wake 105.0%
L2 51.2% | wake 91.1%
L2 41.2% | wake 74.0%
L2 30.6% | wake 54.8%

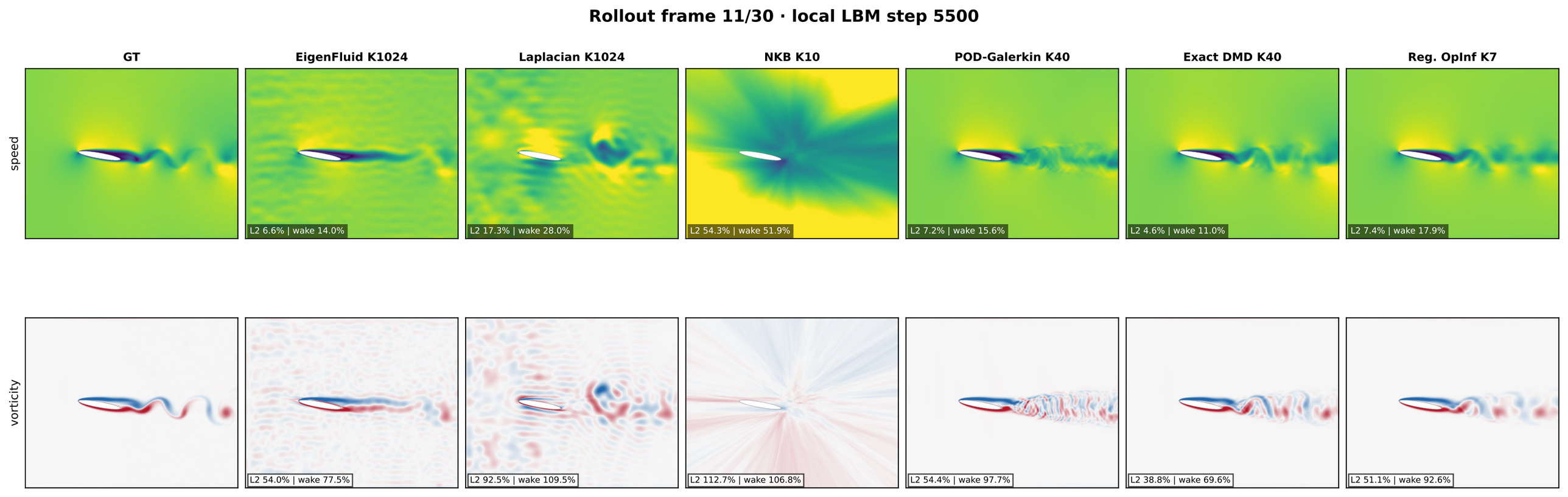

Rollout frame 06/30 · local LBM step 3000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 4.5% | wake 8.4%
L2 14.0% | wake 20.8%
L2 42.1% | wake 48.9%
L2 5.8% | wake 12.0%
L2 5.5% | wake 12.6%
L2 7.9% | wake 18.6%
L2 41.2% | wake 54.2%
L2 90.3% | wake 106.4%
L2 110.6% | wake 105.1%
L2 50.4% | wake 90.5%
L2 39.7% | wake 70.8%
L2 51.7% | wake 93.2%

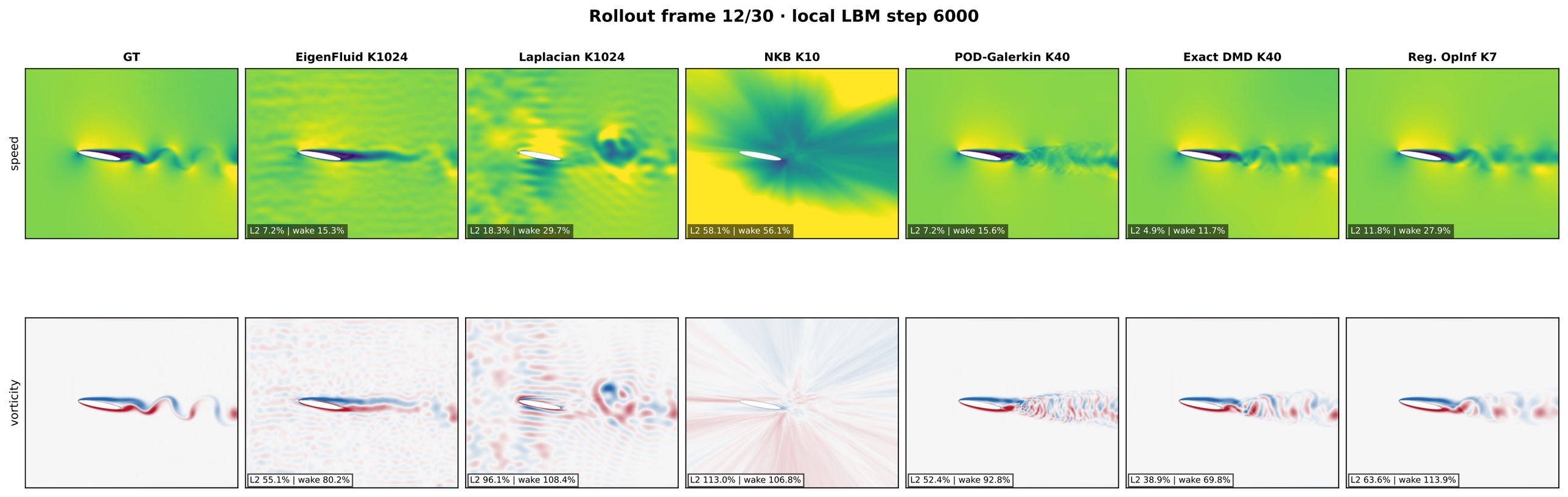

Rollout frame 07/30 · local LBM step 3500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 5.3% | wake 10.0%
L2 15.0% | wake 26.3%
L2 44.9% | wake 51.3%
L2 5.9% | wake 12.9%
L2 4.3% | wake 9.9%
L2 12.3% | wake 28.9%
L2 43.3% | wake 58.3%
L2 89.5% | wake 114.8%
L2 110.7% | wake 104.9%
L2 57.1% | wake 101.2%
L2 37.1% | wake 66.2%
L2 64.2% | wake 114.9%

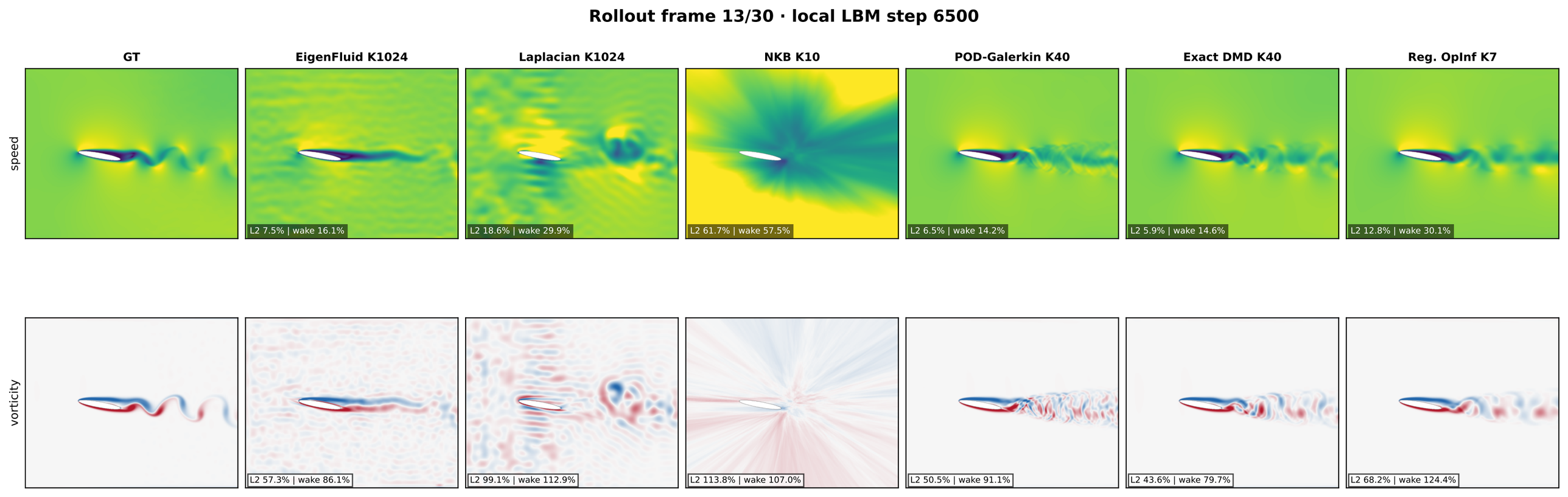

Rollout frame 08/30 · local LBM step 4000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 5.5% | wake 10.6%
L2 15.2% | wake 27.1%
L2 47.3% | wake 51.3%
L2 6.0% | wake 13.4%
L2 5.0% | wake 11.1%
L2 12.8% | wake 29.9%
L2 47.7% | wake 64.5%
L2 88.5% | wake 111.2%
L2 111.2% | wake 105.2%
L2 59.3% | wake 105.4%
L2 38.9% | wake 70.2%
L2 66.0% | wake 119.2%

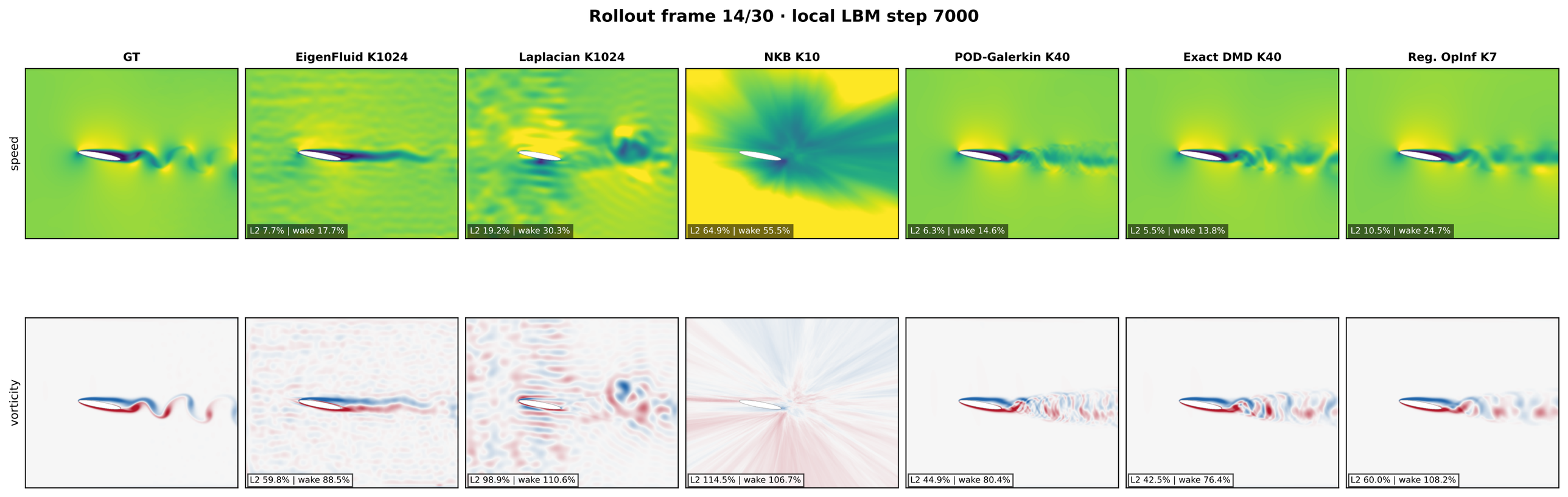

Rollout frame 09/30 · local LBM step 4500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 5.9% | wake 12.3%
L2 15.9% | wake 28.0%
L2 49.4% | wake 49.7%
L2 6.5% | wake 15.6%
L2 4.9% | wake 11.2%
L2 9.8% | wake 22.7%
L2 50.8% | wake 69.7%
L2 89.1% | wake 109.8%
L2 111.7% | wake 105.5%
L2 64.3% | wake 115.1%
L2 41.0% | wake 73.3%
L2 53.8% | wake 96.6%

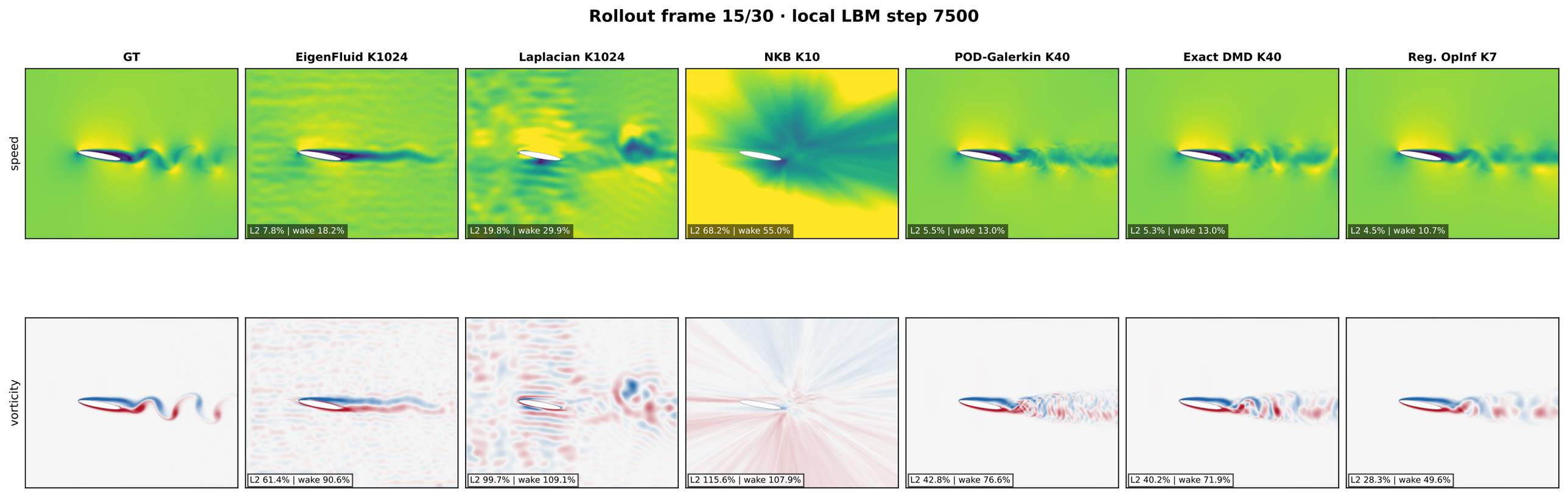

Rollout frame 10/30 · local LBM step 5000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 6.0% | wake 13.5%
L2 16.6% | wake 28.4%
L2 51.4% | wake 49.3%
L2 6.7% | wake 15.6%
L2 4.6% | wake 11.1%
L2 4.6% | wake 10.7%
L2 52.3% | wake 73.2%
L2 90.6% | wake 111.6%
L2 112.4% | wake 106.5%
L2 59.2% | wake 106.2%
L2 40.4% | wake 72.7%
L2 29.4% | wake 52.1%

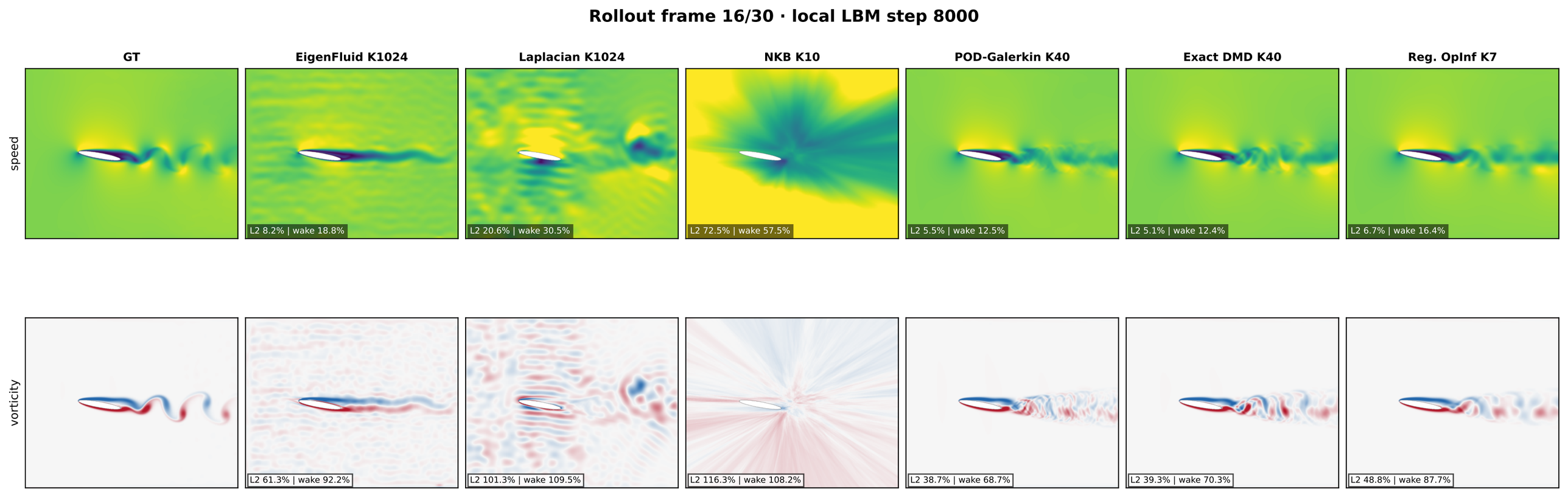

Rollout frame 11/30 · local LBM step 5500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 6.6% | wake 14.0%
L2 17.3% | wake 28.0%
L2 54.3% | wake 51.9%
L2 7.2% | wake 15.6%
L2 4.6% | wake 11.0%
L2 7.4% | wake 17.9%
L2 54.0% | wake 77.5%
L2 92.5% | wake 109.5%
L2 112.7% | wake 106.8%
L2 54.4% | wake 97.7%
L2 38.8% | wake 69.6%
L2 51.1% | wake 92.6%

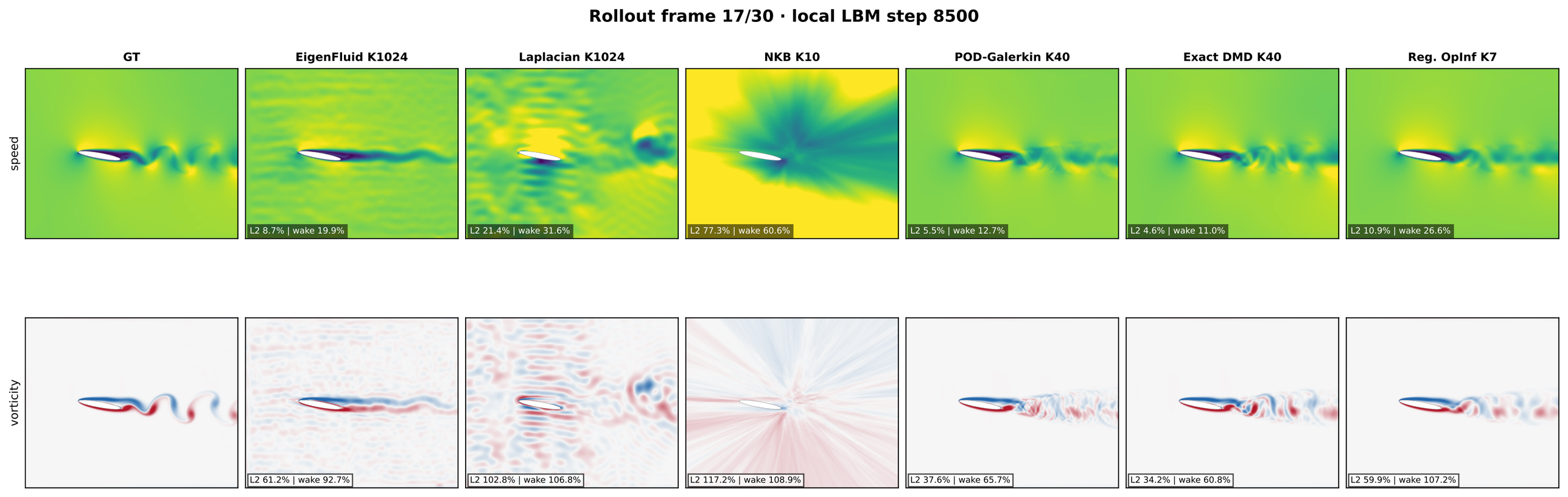

Rollout frame 12/30 · local LBM step 6000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 7.2% | wake 15.3%
L2 18.3% | wake 29.7%
L2 58.1% | wake 56.1%
L2 7.2% | wake 15.6%
L2 4.9% | wake 11.7%
L2 11.8% | wake 27.9%
L2 55.1% | wake 80.2%
L2 96.1% | wake 108.4%
L2 113.0% | wake 106.8%
L2 52.4% | wake 92.8%
L2 38.9% | wake 69.8%
L2 63.6% | wake 113.9%

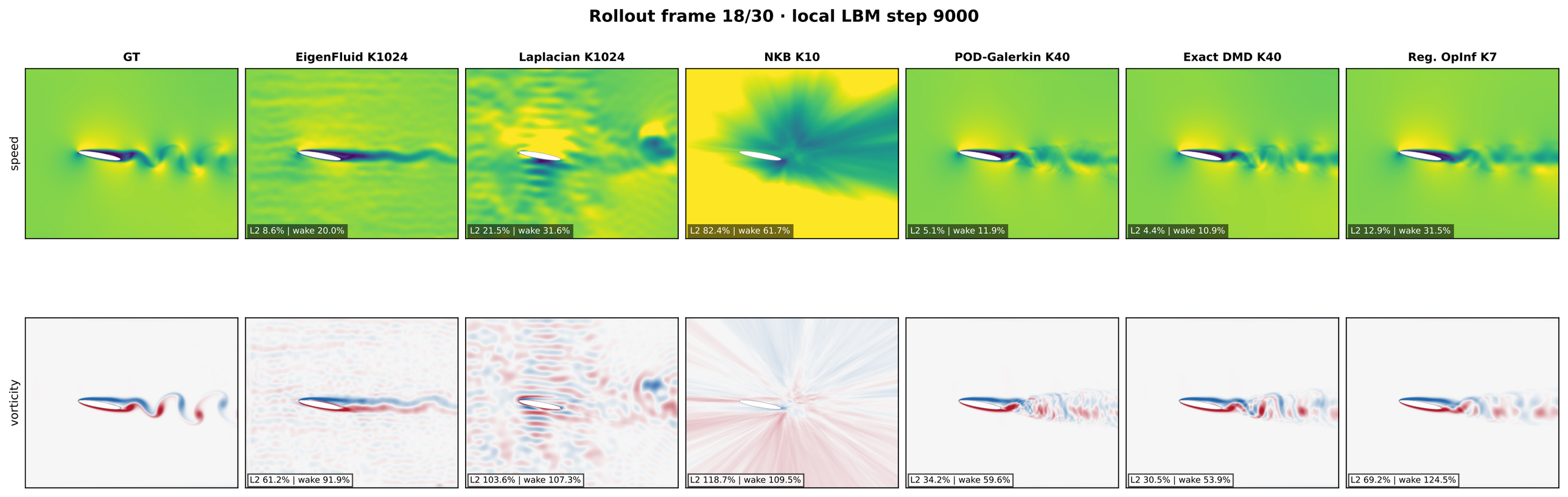

Rollout frame 13/30 · local LBM step 6500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 7.5% | wake 16.1%
L2 18.6% | wake 29.9%
L2 61.7% | wake 57.5%
L2 6.5% | wake 14.2%
L2 5.9% | wake 14.6%
L2 12.8% | wake 30.1%
L2 57.3% | wake 86.1%
L2 99.1% | wake 112.9%
L2 113.8% | wake 107.0%
L2 50.5% | wake 91.1%
L2 43.6% | wake 79.7%
L2 68.2% | wake 124.4%

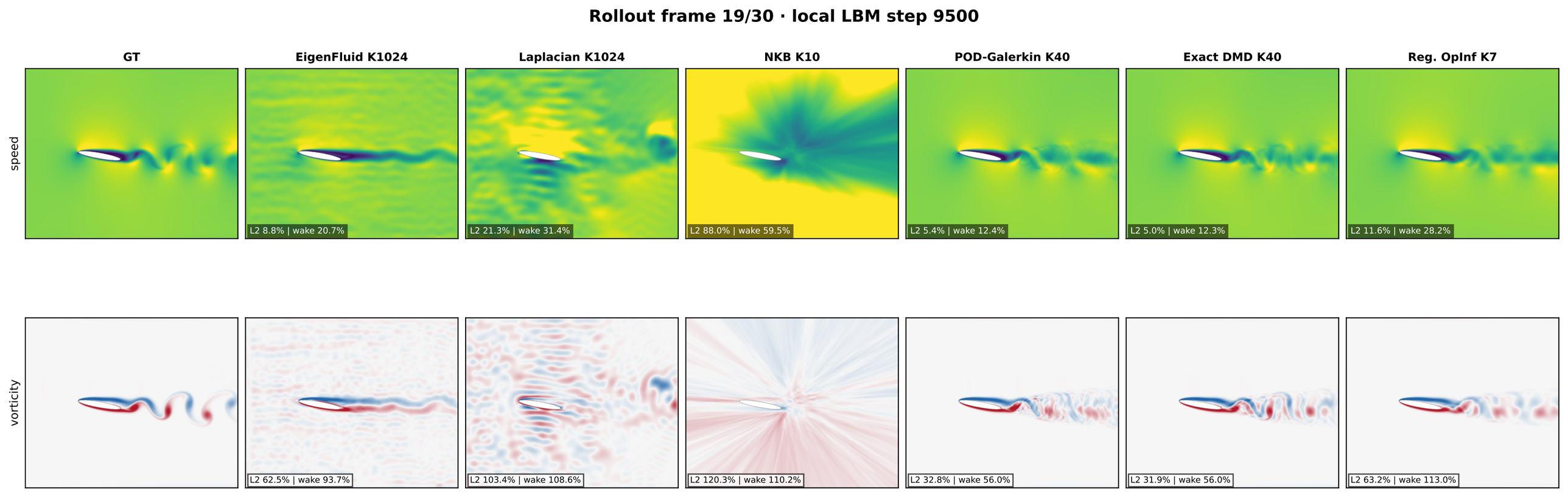

Rollout frame 14/30 · local LBM step 7000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 7.7% | wake 17.7%
L2 19.2% | wake 30.3%
L2 64.9% | wake 55.5%
L2 6.3% | wake 14.6%
L2 5.5% | wake 13.8%
L2 10.5% | wake 24.7%
L2 59.8% | wake 88.5%
L2 98.9% | wake 110.6%
L2 114.5% | wake 106.7%
L2 44.9% | wake 80.4%
L2 42.5% | wake 76.4%
L2 60.0% | wake 108.2%

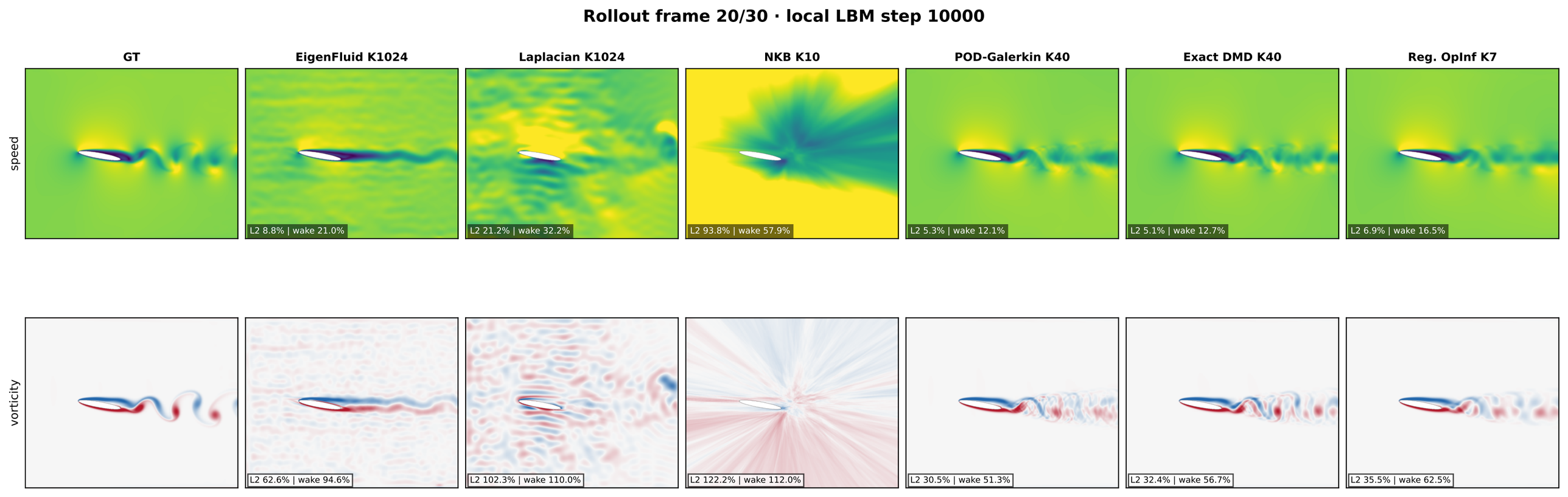

Rollout frame 15/30 · local LBM step 7500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 7.8% | wake 18.2%
L2 19.8% | wake 29.9%
L2 68.2% | wake 55.0%
L2 5.5% | wake 13.0%
L2 5.3% | wake 13.0%
L2 4.5% | wake 10.7%
L2 61.4% | wake 90.6%
L2 99.7% | wake 109.1%
L2 115.6% | wake 107.9%
L2 42.8% | wake 76.6%
L2 40.2% | wake 71.9%
L2 28.3% | wake 49.6%

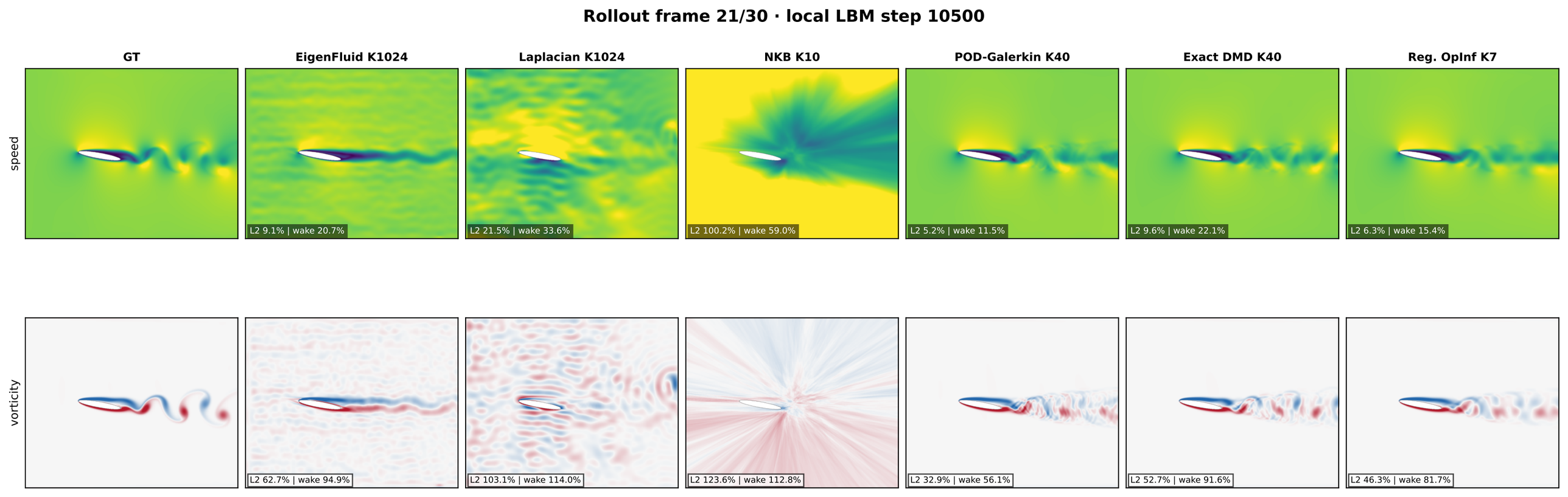

Rollout frame 16/30 · local LBM step 8000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 8.2% | wake 18.8%
L2 20.6% | wake 30.5%
L2 72.5% | wake 57.5%
L2 5.5% | wake 12.5%
L2 5.1% | wake 12.4%
L2 6.7% | wake 16.4%
L2 61.3% | wake 92.2%
L2 101.3% | wake 109.5%
L2 116.3% | wake 108.2%
L2 38.7% | wake 68.7%
L2 39.3% | wake 70.3%
L2 48.8% | wake 87.7%

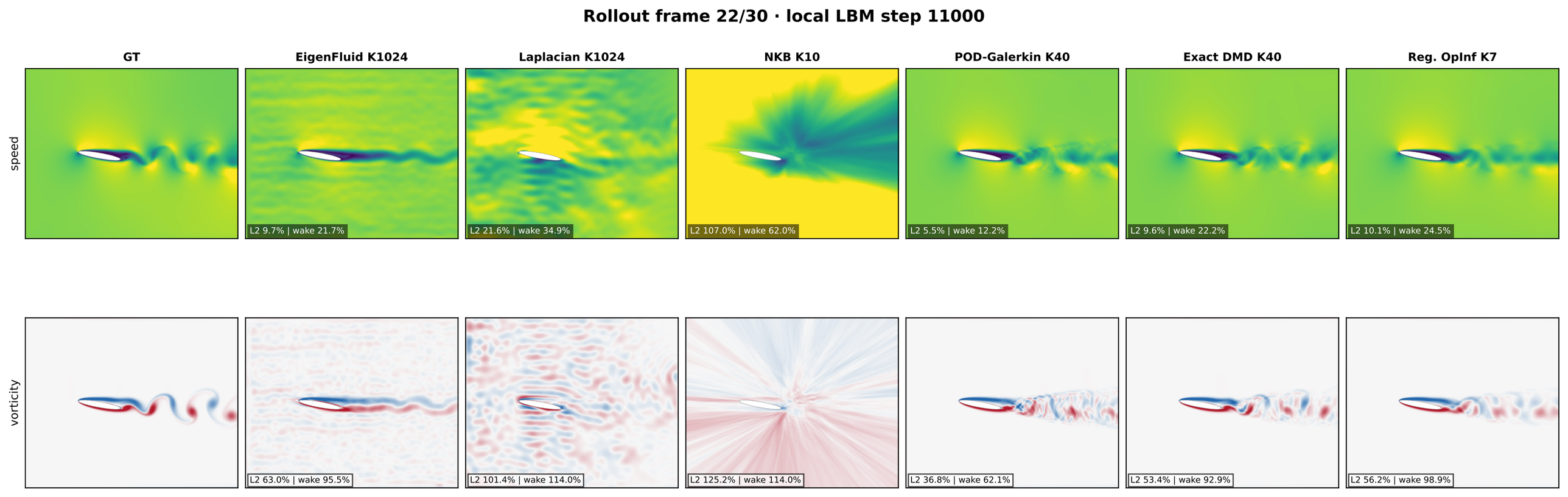

Rollout frame 17/30 · local LBM step 8500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 8.7% | wake 19.9%
L2 21.4% | wake 31.6%
L2 77.3% | wake 60.6%
L2 5.5% | wake 12.7%
L2 4.6% | wake 11.0%
L2 10.9% | wake 26.6%
L2 61.2% | wake 92.7%
L2 102.8% | wake 106.8%
L2 117.2% | wake 108.9%
L2 37.6% | wake 65.7%
L2 34.2% | wake 60.8%
L2 59.9% | wake 107.2%

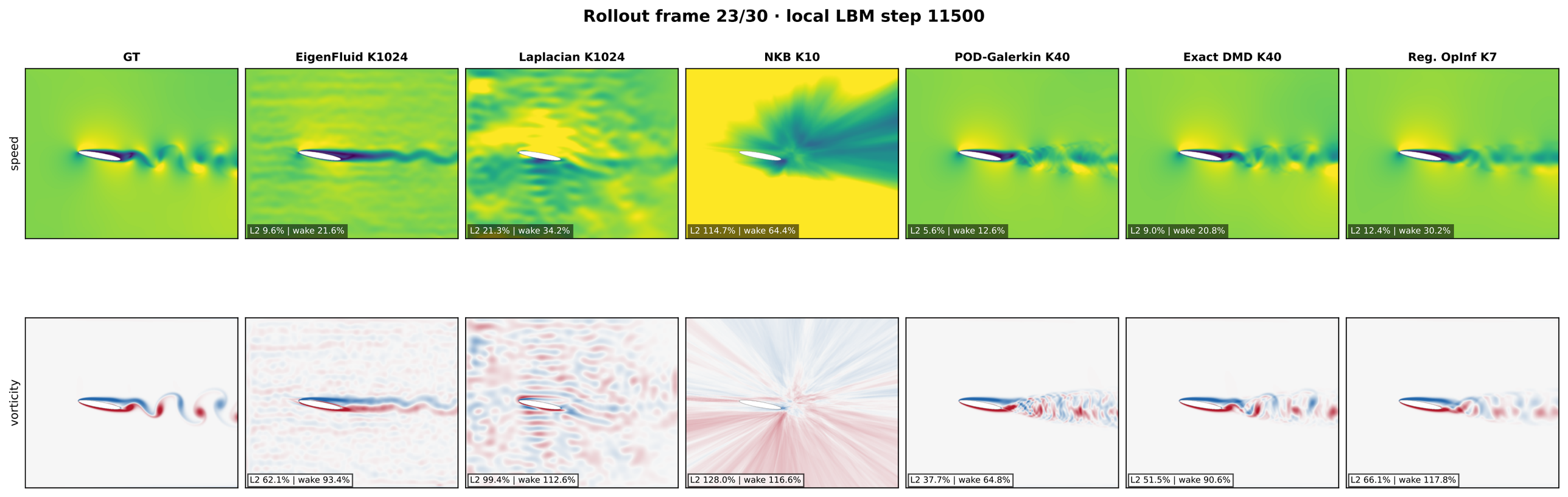

Rollout frame 18/30 · local LBM step 9000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 8.6% | wake 20.0%
L2 21.5% | wake 31.6%
L2 82.4% | wake 61.7%
L2 5.1% | wake 11.9%
L2 4.4% | wake 10.9%
L2 12.9% | wake 31.5%
L2 61.2% | wake 91.9%
L2 103.6% | wake 107.3%
L2 118.7% | wake 109.5%
L2 34.2% | wake 59.6%
L2 30.5% | wake 53.9%
L2 69.2% | wake 124.5%

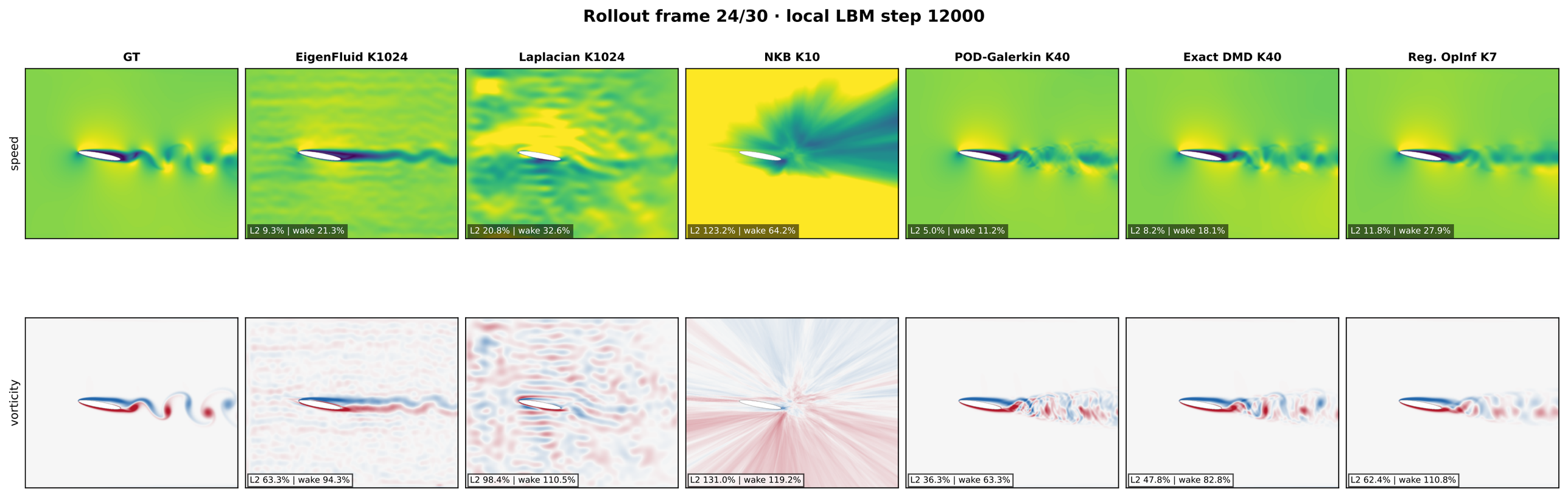

Rollout frame 19/30 · local LBM step 9500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 8.8% | wake 20.7%
L2 21.3% | wake 31.4%
L2 88.0% | wake 59.5%
L2 5.4% | wake 12.4%
L2 5.0% | wake 12.3%
L2 11.6% | wake 28.2%
L2 62.5% | wake 93.7%
L2 103.4% | wake 108.6%
L2 120.3% | wake 110.2%
L2 32.8% | wake 56.0%
L2 31.9% | wake 56.0%
L2 63.2% | wake 113.0%

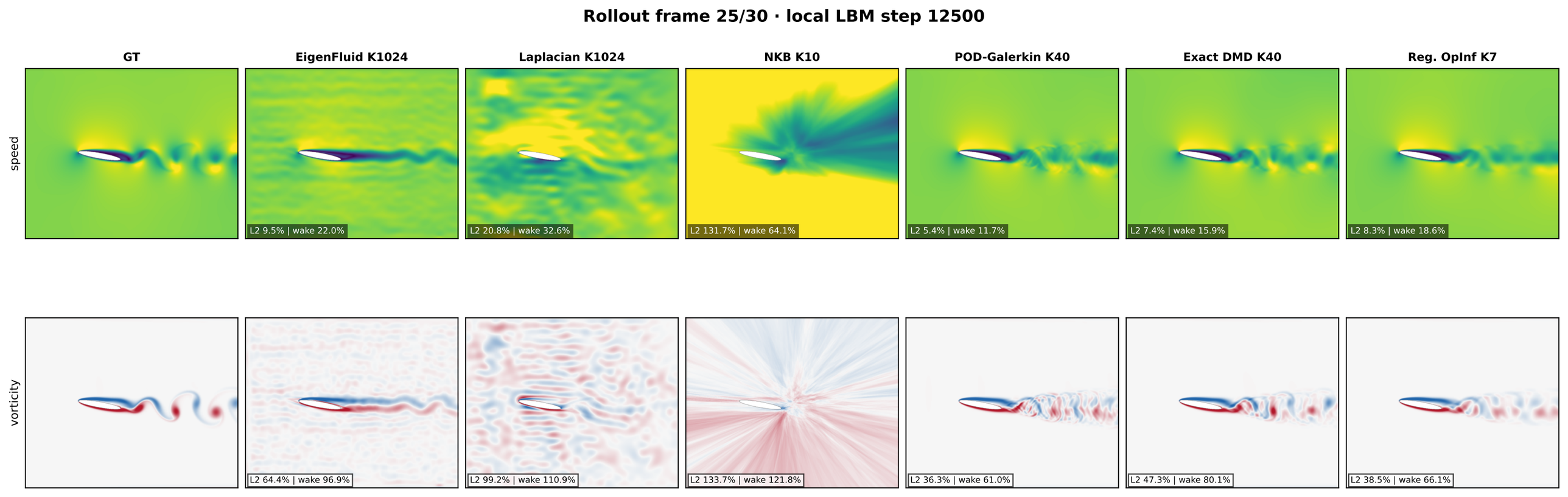

Rollout frame 20/30 · local LBM step 10000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
L2 8.8% | wake 21.0%
L2 21.2% | wake 32.2%
L2 93.8% | wake 57.9%
L2 5.3% | wake 12.1%
L2 5.1% | wake 12.7%
L2 6.9% | wake 16.5%
vorticity
L2 62.6% | wake 94.6%
L2 102.3% | wake 110.0%
L2 122.2% | wake 112.0%
L2 30.5% | wake 51.3%
L2 32.4% | wake 56.7%
L2 35.5% | wake 62.5%

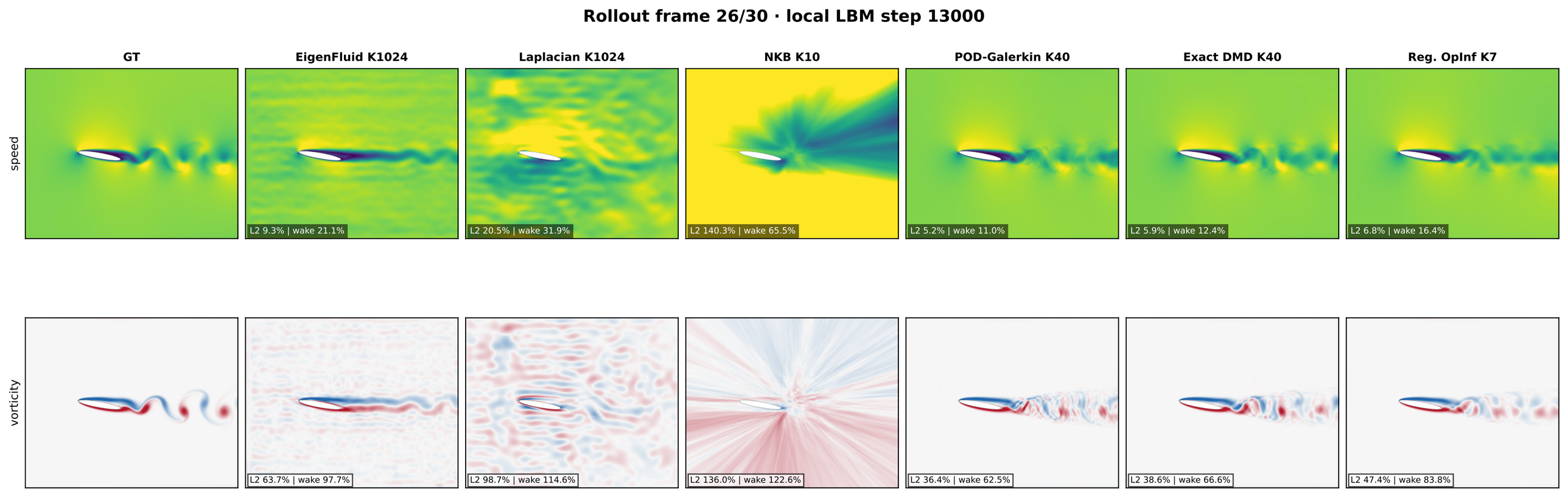

Rollout frame 21/30 · local LBM step 10500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 9.1% | wake 20.7%
L2 21.5% | wake 33.6%
L2 100.2% | wake 59.0%
L2 5.2% | wake 11.5%
L2 9.6% | wake 22.1%
L2 6.3% | wake 15.4%
L2 62.7% | wake 94.9%
L2 103.1% | wake 114.0%
L2 123.6% | wake 112.8%
L2 32.9% | wake 56.1%
L2 52.7% | wake 91.6%
L2 46.3% | wake 81.7%

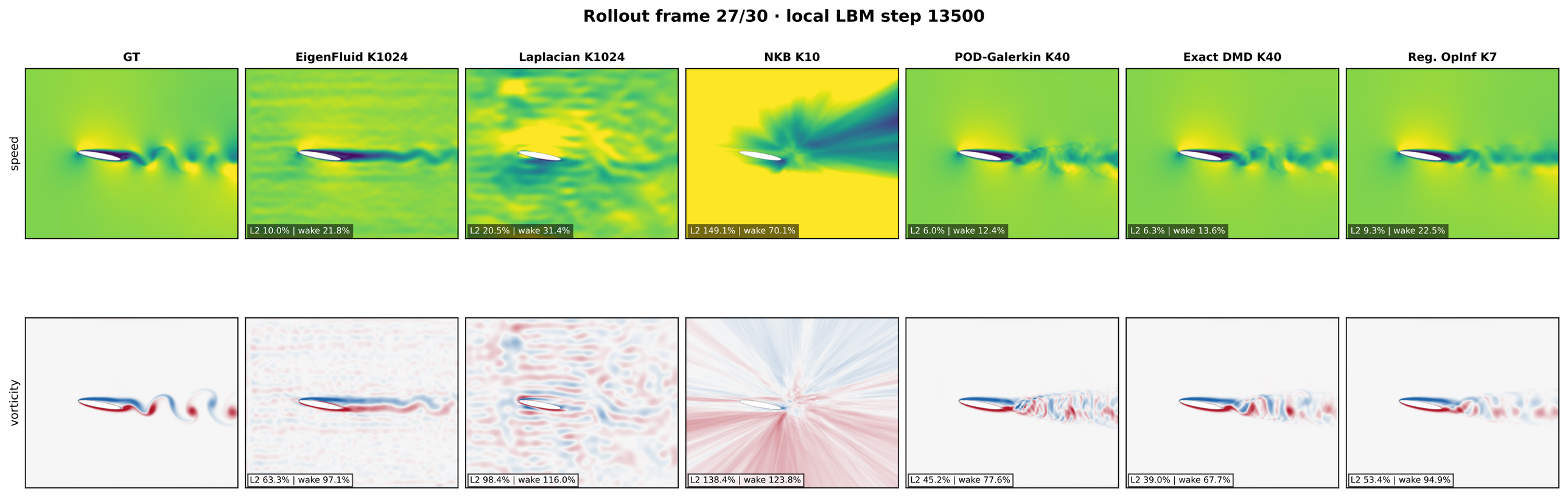

Rollout frame 22/30 · local LBM step 11000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 9.7% | wake 21.7%
L2 21.6% | wake 34.9%
L2 107.0% | wake 62.0%
L2 5.5% | wake 12.2%
L2 9.6% | wake 22.2%
L2 10.1% | wake 24.5%
L2 63.0% | wake 95.5%
L2 101.4% | wake 114.0%
L2 125.2% | wake 114.0%
L2 36.8% | wake 62.1%
L2 53.4% | wake 92.9%
L2 56.2% | wake 98.9%

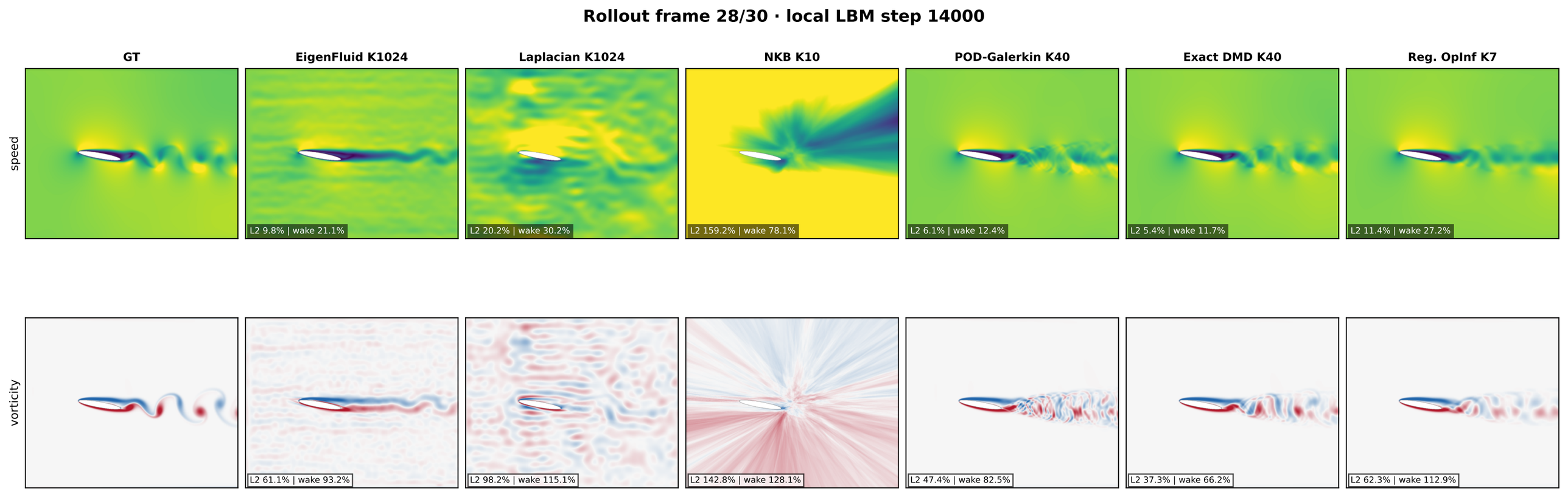

Rollout frame 23/30 · local LBM step 11500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 9.6% | wake 21.6%
L2 21.3% | wake 34.2%
L2 114.7% | wake 64.4%
L2 5.6% | wake 12.6%
L2 9.0% | wake 20.8%
L2 12.4% | wake 30.2%
L2 62.1% | wake 93.4%
L2 99.4% | wake 112.6%
L2 128.0% | wake 116.6%
L2 37.7% | wake 64.8%
L2 51.5% | wake 90.6%
L2 66.1% | wake 117.8%

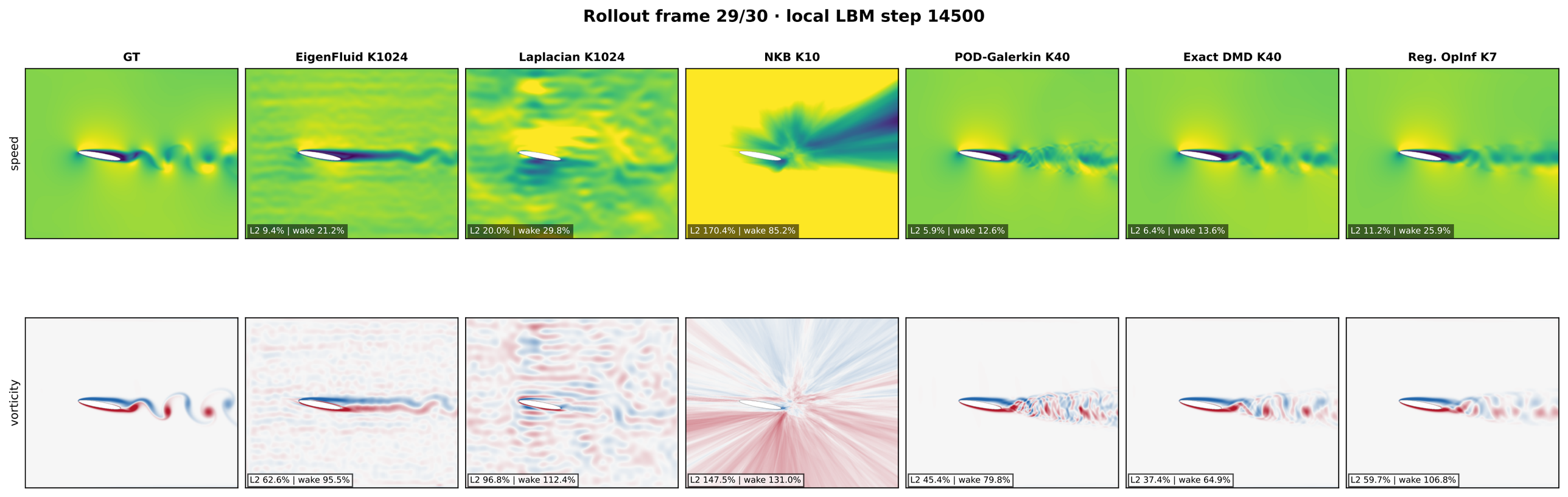

Rollout frame 24/30 · local LBM step 12000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 9.3% | wake 21.3%
L2 20.8% | wake 32.6%
L2 123.2% | wake 64.2%
L2 5.0% | wake 11.2%
L2 8.2% | wake 18.1%
L2 11.8% | wake 27.9%
L2 63.3% | wake 94.3%
L2 98.4% | wake 110.5%
L2 131.0% | wake 119.2%
L2 36.3% | wake 63.3%
L2 47.8% | wake 82.8%
L2 62.4% | wake 110.8%

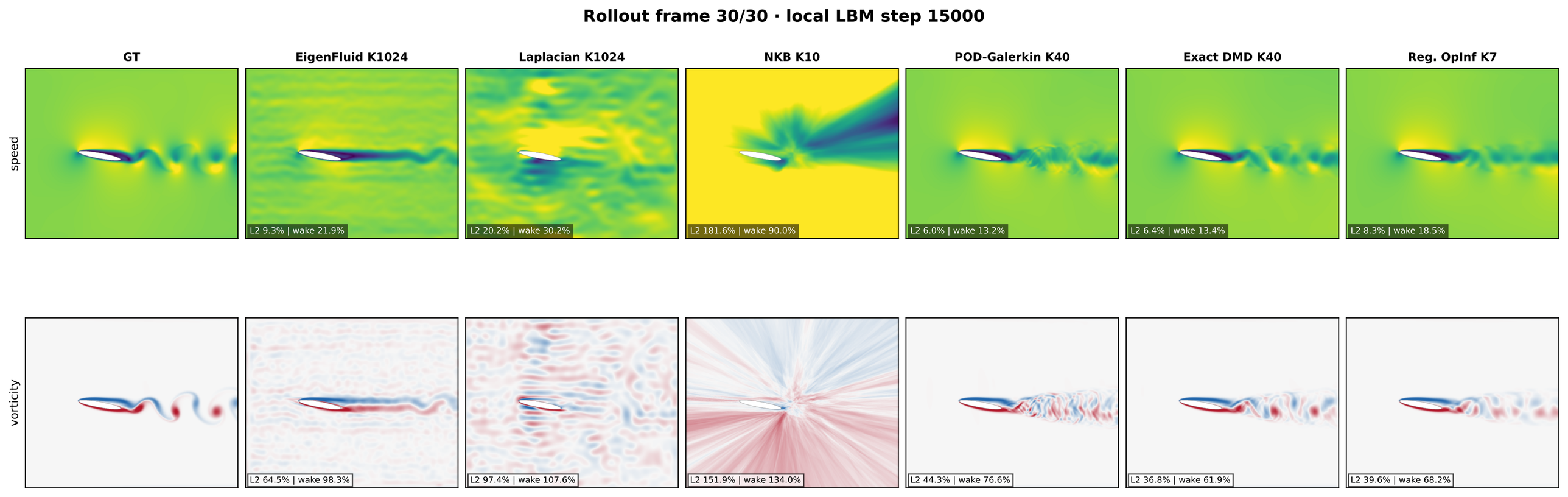

Rollout frame 25/30 · local LBM step 12500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 9.5% | wake 22.0%
L2 20.8% | wake 32.6%
L2 131.7% | wake 64.1%
L2 5.4% | wake 11.7%
L2 7.4% | wake 15.9%
L2 8.3% | wake 18.6%
L2 64.4% | wake 96.9%
L2 99.2% | wake 110.9%
L2 133.7% | wake 121.8%
L2 36.3% | wake 61.0%
L2 47.3% | wake 80.1%
L2 38.5% | wake 66.1%


Rollout frame 26/30 · local LBM step 13000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 9.3% | wake 21.1%
L2 20.5% | wake 31.9%
L2 140.3% | wake 65.5%
L2 5.2% | wake 11.0%
L2 5.9% | wake 12.4%
L2 6.8% | wake 16.4%
L2 63.7% | wake 97.7%
L2 98.7% | wake 114.6%
L2 136.0% | wake 122.6%
L2 36.4% | wake 62.5%
L2 38.6% | wake 66.6%
L2 47.4% | wake 83.8%


Rollout frame 27/30 · local LBM step 13500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 10.0% | wake 21.8%
L2 20.5% | wake 31.4%
L2 149.1% | wake 70.1%
L2 6.0% | wake 12.4%
L2 6.3% | wake 13.6%
L2 9.3% | wake 22.5%
L2 63.3% | wake 97.1%
L2 98.4% | wake 116.0%
L2 138.4% | wake 123.8%
L2 45.2% | wake 77.6%
L2 39.0% | wake 67.7%
L2 53.4% | wake 94.9%


Rollout frame 28/30 · local LBM step 14000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 9.8% | wake 21.1%
L2 20.2% | wake 30.2%
L2 159.2% | wake 78.1%
L2 6.1% | wake 12.4%
L2 5.4% | wake 11.7%
L2 11.4% | wake 27.2%
L2 61.1% | wake 93.2%
L2 98.2% | wake 115.1%
L2 142.8% | wake 128.1%
L2 47.4% | wake 82.5%
L2 37.3% | wake 66.2%
L2 62.3% | wake 112.9%


Rollout frame 29/30 · local LBM step 14500
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 9.4% | wake 21.2%
L2 20.0% | wake 29.8%
L2 170.4% | wake 85.2%
L2 5.9% | wake 12.6%
L2 6.4% | wake 13.6%
L2 11.2% | wake 25.9%
L2 62.6% | wake 95.5%
L2 96.8% | wake 112.4%
L2 147.5% | wake 131.0%
L2 45.4% | wake 79.8%
L2 37.4% | wake 64.9%
L2 59.7% | wake 106.8%


Rollout frame 30/30 · local LBM step 15000
GT
EigenFluid K1024
Laplacian K1024
NKB K10
POD-Galerkin K40
Exact DMD K40
Reg. OpInf K7
speed
vorticity
L2 9.3% | wake 21.9%
L2 20.2% | wake 30.2%
L2 181.6% | wake 90.0%
L2 6.0% | wake 13.2%
L2 6.4% | wake 13.4%
L2 8.3% | wake 18.5%
L2 64.5% | wake 98.3%
L2 97.4% | wake 107.6%
L2 151.9% | wake 134.0%
L2 44.3% | wake 76.6%
L2 36.8% | wake 61.9%
L2 39.6% | wake 68.2%